2022-26

Full Marks : 70

Time : 3 hours

Answer from **all** the Sections as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

SECTION—A

Answer any *four* questions of the following: 10×4

1. A coin is tossed 8 times. Find by using binomial distribution the probability of getting

   (*a*) At least 3 heads.

   (*b*) Exactly 3 heads.

   (*c*) At most 3 heads.

2. State and prove Baye's theorem.

3. Find the mean, Median and Mode of the following data:

Marks obtained	No. of students
Less than 10	5
Less than 20	9
Less than 30	17
Less than 40	29
Less than 50	45
Less than 60	60
Less than 70	70
Less than 80	78
Less than 90	83
Less than 100	85

(*Continued*)

4. Calculate the correlation co-efficient for the heights of fathers (X) and sons (Y):

X	65	66	67	68	69	70	72	67
Y	67	68	65	72	72	69	71	68

5. Define regression co-efficient and discuss its properties.

6. Find range, standard deviation and variance for the following data:

87 99 75 87 94 75 35 88 87

7. Define random variable and distribution function of discrete random variables.

8. Calculate the chi-square (x^2) test from the following data:

	Hb %		
	Above normal	Below normal	Total
Above normal	85	75	160
Below normal	165	175	340
Total	250	250	500

SECTION—B

Answer *all* questions: 3 × 10

9. Define Event and Sample space.

10. Define Chi-Square.

11. Define Quartile Deviations.

12. Write merit and demerit of mode.

13. What is Regression Co-efficient ?

14. Define Skewness and Kurtosis.

15. Define Karl Pearson's Co-efficient of Correlation.

16. Define Test of goodness of fit.

17. Write Hypothesis testing procedure.

18. Define Binomial distribution.

2023(New)

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Sections as directed.

Section – A

Answer any **four** questions of the following : 10×4 = 40

1. Define sample space and find it for tossing of two coins.
2. Define random variable and distribution function of discrete random variables.
3. Define Poisson distribution.
4. Define correlation coefficient and prove that its lies between – 1 and 1.
5. Define regression coefficient.

6. Prove that , if one regression coefficient is greater than unity, then the other must be less than unity.

7. Find range and standard deviation for the data

87, 99, 75, 87, 94, 75, 35, 88, 87..

8. Find the correlation coefficient by Karl Pearson's method :

x	y
10	3
17	8
12	12
13	8
14	6
15	3

Section – B

Answer **all** questions of the following : $3 \times 10 = 30$

9. Prove that $P(E) + P(E') = 1$.

10 Prove that $A \subseteq B \Rightarrow P(A) \leq P(B)$.

11. A random variable x has the following probability values :

x	P(x)
0	0
1	k
2	2k
3	3k
4	4k

then, find the value of k.

12. Prove that the mean and variance of a Poisson distribution are equal.

13. Calculate the coefficient of correlation between the value of x and y :

X	Y
78	125
89	137
97	156
69	112
59	107
78	136
68	123
61	108

14. Obtain the equations of two lines of regression for the following data :

X	Y
65	67
66	68
6	65
67	68
68	72
69	72
70	69
72	71

15. Explain the properties and application of Chi-square distribution.

16. For the following set of points {(–2,–1), (1,1), (3, 2)} Find the least sqaure regression line.

17. Write short note on Hypothesis testing procedure.

18. Write short note on the Test of goodness of fit.

Section – B

9. Answer any **ten** questions of the following : 3×10 = 30

(a) Define event, sample space and statistics.

(b) What is Poison's distribution ?

(c) Define Quartile Deviations.

(d) Define Chi-square.

(e) What is Regression Coefficient ?

(f) What is moments ?

(g) Discuss about variance.

(h) What is rank ?

(i) Write difference between M.D. and S.D.

(j) Write relation among A.M, G.M. and H.M. with examples.

(k) Write about T-distribution.

(l) Define conditional probability.

(m) Write merit and demerit of Mode.

(n) Write Karl Pearson's coefficient of correlation.

(o) Give an example of normal distribution.

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2022

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Sections as directed

Section – A

Answer any **four** questions of the following : 10×4 = 40

1. State and Prove Bay's theorem.

2. A coin is tossed 8 times. F... by using binomial distribution the probability of getting
(a) At least 3 heads (b) Exactly 3 heads
(c) At most 3 heads.

(Turn over)

3. Three persons A, B and C throw a die in succession till one gets a six and wins the game. Find their respective probabilities of winning.

4. Find the Mean, Median and Mode of the following data :

Marks obtained	No of students
Less than 10	5
Less than 20	9
Less than 30	17
Less than 40	29
Less than 50	45
Less than 60	60
Less than 70	70
Less than 80	78
Less than 90	83
Less than 100	85

5. Discuss Skewness and Kurtosis with example.

6. Show that correlation coefficient is independent of change of origin and scale.

7. Obtain the equation of the line of regression for the following data :

X	Y
65	67
66	68
67	65
67	68
68	72
69	72
70	69
72	71

8. Calculated the Chi-Square (χ^2) test from the following data :

	Hb%		
	Above normal	Below normal	Total
Above normal	85	75	160
Below normal	165	175	340
Total	250	250	500

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2018

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The questions are of equal value.

*Answer any **five** questions.*

1. (a) Define sample space and find it for the following experiments :

   (i) Tossing of two dice

   (ii) Tossing of two coins

   (b) Prove that :

   (i) $P(E) + P(E') = 1$

   (ii) $A \subseteq B \Rightarrow P(A) \leq P(B)$

2. (a) Define random variable and distribution function of discrete random variables.

(b) A random variable x has the following probability values :

x	P(x)
0	0
1	k
2	2k
3	2k
4	3k

(i) Find the value of k.

(ii) Determine the distribution function of x.

3. (a) Define Poisson distribution.

(b) Prove that mean and varience of a Poisson distribution are equal.

4. Find ranges standard deviation and variance for the data, 87, 99, 75, 87, 94, 75, 35, 88, 87.

5. (a) Define correlation coeeficient and prove that it sies between –1 to +1.

(b) Calculate the coefficient of correlation between the value of x and y given below :

x	Y
78	125
89	137
97	156
69	112
59	107
79	136
68	123
61	108

6. (a) Define regression coefficients and prove that if one regression coefficient is greater than unity, then , the other must be less than unity.

(b) Obtain the equations of two lines of regression for the following data :

x	Y
65	67
66	68
67	65

x	Y
67	68
68	72
69	72
70	69
72	71

7. Define chi-square distribution and explain its properties and applications.

8. For the following set of points {(–2, –1), (1, 1), (3, 2)} :

   (a) Find the least square regression line.

   (b) Plot the given points and the regression line in the same rectangular system of axes.

9. Write notes on the following :

   (a) Hypothesis testing procedure

   (b) Test of goodness of fit

